Equation of Lines in R^3

Knowledge/ Understanding

2. Find a direction vector for a line
   a. parallel to $\vec{r} = (7, -9, 3) + t(-4, 2, -5)$
   b. through (0, 6, 3) and (7, 4, 6)
   c. parallel to $-x = \frac{y-3}{2} = \frac{z}{4}$

3. Give the coordinates of two points on each of the following lines.
   a. $\vec{r} = (1, 1, 2) + t(3, -1, -1)$
   b. $x = 4 - 2t, y = -2 + 5t, z = 5 + 4t$
   c. $\frac{x-4}{3} = \frac{y+5}{4} = \frac{z+1}{-1}$

Knowledge/ Understanding

4. For each of the following, find vector, parametric, and, if possible, symmetric equations of the line that passes through P_0 and has direction vector $\vec{d}$.
   a. $P_0(2, 4, 6), \vec{d} = (-1, -3, 2)$
   b. $P_0(0, 0, -5), \vec{d} = (-1, 4, 1)$
   c. $P_0(1, 0, 0), \vec{d} = (0, 0, -1)$

Application

6. a. Which of the following points lies on the line $x = 2t, y = 3 + t, z = 1 + t$?
   $P(2, 4, 2)$ $Q(-2, 2, 1)$ $R(4, 5, 2)$ $S(6, 6, 2)$
   b. If the point $(a, b, -3)$ lies on the line, find the values of a and b.

Part B

7. Find parametric equations for the line that passes through the point $(0, -1, 1)$ and the midpoint of the line segment from $(2, 3, -2)$ to $(4, -1, 5)$.

Thinking/Inquiry/ Problem Solving

8. Find symmetric equations for the line through the origin that is parallel to the line through the points $(4, 3, 1)$ and $(-2, -4, 3)$.

9. For each of the following pairs of equations, determine whether they represent the same line, parallel lines, or neither of these.
   a. $\vec{r} = (1, 0, 3) + s(3, -6, 3)$ and $\vec{r} = (2, -2, 5) + t(2, -4, 2)$
   b. $\vec{r} = (2, -1, 4) + s(3, 0, 6)$ and $\vec{r} = (-3, 0, 1) + t(2, 0, 2)$
   c. $\vec{r} = (1, -1, 1) + s(6, 2, 0)$ and $\vec{r} = (-5, -3, 1) + t(-9, -3, 0)$

Answers

2. a. $(4, -2, 5)$ **b.** $(7, -2, 3)$ **c.** $(-1, 2, 4)$
3. a. $(4, 0, 1), (-5, 3, 4)$ **b.** $(4, -2, 5), (2, 3, 9)$
c. $(4, -5, -1), (7, -1, -2)$ **4. a.** $\vec{r} = (2, 4, 6) + t(1, 3, -2)$;
$x = 2 + t, y = 4 + 3t, z = 6 - 2t$;
$\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{-2}$
b. $\vec{r} = (0, 0, -5) + t(1, -4, -1)$; $x = t, y = -4t, z = -5 - t$;
$\frac{x}{1} = \frac{y}{-4} = \frac{z+5}{-1}$ **c.** $\vec{r} = (1, 0, 0) + t(0, 0, -1)$;
$x = 1, y = 0, z = -t$ **5.** $(-20, 10, -27), (-14, 8, -17)$,
$(-8, 6, -7), (-2, 4, 3), (4, 2, 13), (10, 0, 23), (16, -2, 33)$
6. a. $P(2, 4, 2)$ **b.** $a = -8, b = -1$ **7.** $x = 6t, y = -1 + 4t$,
$z = 1 + t$ **8.** $\frac{x}{6} = \frac{y}{7} = \frac{z}{-2}$ **9. a.** parallel **b.** neither **c.** same